Chapter 9. Comparing Tests

Power, most powerful tests, Neyman–Pearson, nonparametric tests and their efficiency

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9.1 Criteria for comparing tests

Simplest case: **simple** hypotheses, $\Omega = \{\theta_0, \theta_1\}$,

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_1 : \theta = \theta_1.$$

A test ϕ has errors $E_{\theta_0}\phi(\mathbf{X})$ (type I) and $E_{\theta_1}[1-\phi(\mathbf{X})]$ (type II); reducing one increases the other, so combine them:

- **maximum error**: $\max\{E_{\theta_0}\phi, E_{\theta_1}(1-\phi)\}$;
- **Bayes error**: $\pi_0 E_{\theta_0}\phi + \pi_1 E_{\theta_1}(1-\phi)$, priors $\pi_0, \pi_1 > 0$, $\pi_0 + \pi_1 = 1$.

Let $p_\theta(\mathbf{x}) = \prod f(x_i;\theta)$.

Theorem 9.1 (Bayes test for simple hypotheses)

The test

$$\phi^\pi(\mathbf{x}) = \begin{cases} 1, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) > \pi_0/\pi_1, \\ 0, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) < \pi_0/\pi_1 \end{cases}$$

(arbitrary on equality; positive/0 $= +\infty$) minimizes the Bayes error.

Proof. The Bayes error is

$$\pi_0 \int \phi p_{\theta_0} + \pi_1\Big(1 - \int \phi p_{\theta_1}\Big) = \pi_1 - \int \phi(\mathbf{x})\{\pi_1 p_{\theta_1}(\mathbf{x}) - \pi_0 p_{\theta_0}(\mathbf{x})\}\, d\mathbf{x}.$$

minimized by $\phi = 1$ where the brace is positive and 0 where negative. ∎

Example 9.1

$N(\mu, 1)$, $H_0 : \mu = \mu_0$ vs $H_1 : \mu = \mu_1$:

$$\frac{p_{\mu_1}(\mathbf{x})}{p_{\mu_0}(\mathbf{x})} = \exp\Big\{n(\mu_1 - \mu_0)\Big(\bar{x} - \frac{\mu_0 + \mu_1}{2}\Big)\Big\}.$$

For $\mu_1 > \mu_0$, reject when $\bar{x} > \frac{\mu_0+\mu_1}{2} + \frac{\log(\pi_0/\pi_1)}{n(\mu_1-\mu_0)}$ (symmetrically for $\mu_1 < \mu_0$); equal priors give the midpoint.

These criteria are symmetric in H_0, H_1; Chapter 7 instead bounds the type I error by α and maximizes the power $\gamma_\phi(\theta_1) = E_{\theta_1}\phi$.

Definition (Most powerful test)

For simple H_0 and H_1, a test ϕ^{MP} is a **most powerful (MP) test** of level α if (i) $E_{\theta_0}\phi^{MP} \le \alpha$, and (ii) $E_{\theta_1}\phi^{MP} \ge E_{\theta_1}\phi$ for every ϕ with $E_{\theta_0}\phi \le \alpha$.

Theorem 9.2 (Neyman–Pearson lemma)

A test of the form

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) > k, \\ \gamma, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) = k, \\ 0, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) < k, \end{cases} \qquad k \ge 0,\ 0 \le \gamma \le 1,$$

with size $E_{\theta_0}\phi^* = \alpha$ is a most powerful level-α test.

Proof. For any ϕ, $E_{\theta_1}\phi - kE_{\theta_0}\phi = \int \phi\,(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x}$, so

$$E_{\theta_1}\phi - kE_{\theta_0}\phi = \int \phi\,(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x}.$$

$$\{E_{\theta_1}\phi^* - kE_{\theta_0}\phi^*\} - \{E_{\theta_1}\phi - kE_{\theta_0}\phi\} = \int (\phi^* - \phi)(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x} \geq 0,$$

since $\phi^* = 1 \geq \phi$ where $p_{\theta_1} > kp_{\theta_0}$ and $\phi^* = 0 \leq \phi$ where $p_{\theta_1} < kp_{\theta_0}$. So if $E_{\theta_0}\phi \leq \alpha$,

$$E_{\theta_1}\phi^* - E_{\theta_1}\phi \geq k\{E_{\theta_0}\phi^* - E_{\theta_0}\phi\} = k\{\alpha - E_{\theta_0}\phi\} \geq 0.$$

■

Such k, γ always exist, and every MP test has this form a.s.; γ may depend on $\mathbf{x}$.

Example 9.2

Poisson(θ), $n = 50$, $H_0 : \theta = 0.1$ vs $H_1 : \theta = 0.05$: the ratio $e^{2.5}(1/2)^{\sum x_i}$ decreases in $S = \sum x_i$, so reject for $S \leq c - 1$, randomize at $S = c$, with $E_{0.1}\phi^* = 0.05$: $c = 2$, $\gamma = 0.114$ (Example 7.3).

9.2 Uniformly most powerful tests

Motivation. With monotone likelihood ratio, the Neyman–Pearson test is best against all one-sided alternatives at once.

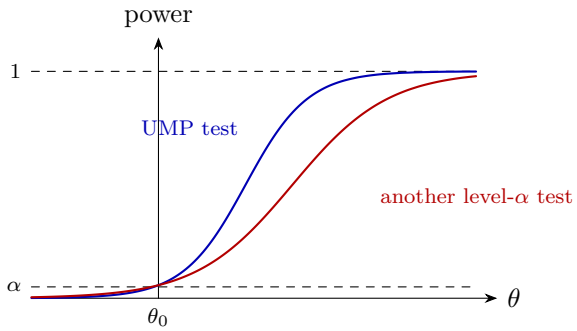


Figure. *Power functions for $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$: both size α; the UMP test has larger power at every $\theta > \theta_0$.*

For composite hypotheses we want large power at every $\theta_1 \in \Omega_1$.

Definition (UMP test)

ϕ^{UMP} is a **uniformly most powerful (UMP) test** of level α if (i) $\max_{\theta\in\Omega_0} E_\theta\phi^{UMP} \leq \alpha$, and (ii) $E_{\theta_1}\phi^{UMP} \geq E_{\theta_1}\phi$ for all $\theta_1 \in \Omega_1$ and every ϕ with $\max_{\Omega_0} E_\theta\phi \leq \alpha$.

Example 9.3

$N(\mu,1)$, $H_0:\mu=\mu_0$ vs $H_1:\mu>\mu_0$. For each $\mu_1>\mu_0$ the ratio increases in $\bar{x}$ (Example 9.1), so the MP test is $\bar{x}\geq\mu_0+z_\alpha/\sqrt{n}$, free of μ_1: UMP,

$$\phi^*(\mathbf{x})=\mathbf{1}\{\bar{x}-\mu_0\geq z_\alpha/\sqrt{n}\}.$$

Example 9.4

It is also UMP for $H_0^-:\mu\leq\mu_0$: its power $P(Z\geq\sqrt{n}(\mu_0-\mu)+z_\alpha)$ increases in μ, so it has level α for H_0^-, and every level-α test for H_0^- has level α at μ_0.

Theorem 9.3 (UMP one-sided tests in exponential families)

Let $f(x;\theta)=\exp\{g(\theta)T(x)-B(\theta)+S(x)\}$, $x\in\mathcal{X}$, $\theta\in\Omega\subseteq\mathbb{R}$, with g increasing. For $H_0:\theta\leq\theta_0$ vs $H_1:\theta>\theta_0$, the test

$$\phi^*(\mathbf{x})=\begin{cases}1, & \sum T(x_i)>c,\\ \gamma, & \sum T(x_i)=c,\\ 0, & \sum T(x_i)<c,\end{cases}\qquad E_{\theta_0}\phi^*=\alpha,$$

is UMP of level α. (For $H_0:\theta\geq\theta_0$ vs $H_1:\theta<\theta_0$, reverse the inequalities.)

Proof.

Step 1. For $\theta_1>\theta_0$,

$$\frac{p_{\theta_1}(\mathbf{x})}{p_{\theta_0}(\mathbf{x})}=\exp\Big\{(g(\theta_1)-g(\theta_0))\sum T(x_i)-n(B(\theta_1)-B(\theta_0))\Big\}$$

increases in $\sum T(x_i)$: ϕ^* is the Neyman–Pearson test, free of θ_1, so UMP for $H_0:\theta=\theta_0$.

Step 2. $E_\theta\phi^*$ is nondecreasing: for $\theta'<\theta''$ and $\alpha'=E_{\theta'}\phi^*$, ϕ^* is MP at level α' for θ' vs θ'', and beats the trivial $\phi\equiv\alpha'$: $E_{\theta''}\phi^*\geq\alpha'$.

Step 3. So $\max_{\theta\leq\theta_0}E_\theta\phi^*=\alpha$, and any level-$\alpha$ test of $H_0:\theta\leq\theta_0$ has level α at θ_0; apply Step 1. ■

Example 9.5

(a) Poisson: $g=\log$ increasing; for $H_0:\theta\geq0.1$ vs $H_1:\theta<0.1$, $n=50$, the UMP level-0.05 test is that of Example 9.2.
(b) Exponential: $g(\theta)=-1/\theta$ increasing; for $H_0:\theta\leq\theta_0$ reject when $\bar{x}/\theta_0\geq\chi^2_\alpha(2n)/(2n)$ (as $2\sum X_i/\theta_0\sim\chi^2(2n)$).

Example 9.6 (No UMP two-sided test)

$N(\mu,1)$, $H_0:\mu=\mu_0$ vs $H_1:\mu\neq\mu_0$. A UMP test would be MP against both $\mu_1<\mu_0$ and $\mu_2>\mu_0$, hence (converse of Neyman–Pearson) both $\mathbf{1}\{\bar{x}\leq c_1\}$ and $\mathbf{1}\{\bar{x}\geq c_2\}$ of size α: impossible.

Example 9.7 (Optimality of the two-sided test)

The LRT $\phi^* = \mathbf{1}\{\sqrt{n}|\bar{x}-\mu_0| \geq z_{\alpha/2}\}$ has power $1-\Phi(z_{\alpha/2}-\delta)+\Phi(-z_{\alpha/2}-\delta)$, $\delta = \sqrt{n}(\mu-\mu_0)$, so (a) $E_{\mu_0}\phi^* = \alpha$ and $\frac{d}{d\mu}E_\mu\phi^*\big|_{\mu_0} = 0$. Claim (b): among tests with these two properties, ϕ^* is most powerful at every $\mu_1 \neq \mu_0$.
With $\dot{p}_{\mu_0} = \partial_\mu p_\mu|_{\mu_0}$, $\frac{d}{d\mu}E_\mu\phi|_{\mu_0} = E_{\mu_0}[\phi\,\dot{p}_{\mu_0}/p_{\mu_0}]$, and for constants k_1, k_2

$$E_{\mu_1}\phi - k_1 E_{\mu_0}\phi - k_2 E_{\mu_0}\Big[\phi\frac{\dot{p}_{\mu_0}}{p_{\mu_0}}\Big] = \int \phi\,\{p_{\mu_1} - k_1 p_{\mu_0} - k_2\dot{p}_{\mu_0}\}\,d\mathbf{x}$$

is maximized by $\phi^{**} = \mathbf{1}\{p_{\mu_1} \geq k_1 p_{\mu_0} + k_2\dot{p}_{\mu_0}\}$. As $\dot{p}_{\mu_0}/p_{\mu_0} = n(\bar{x}-\mu_0)$, this is $\{e^{a(\bar{x}-\mu_0)} \geq b_1 + b_2(\bar{x}-\mu_0)\}$, the complement of an interval (convex exponential above a line). The side conditions make it symmetric: $\phi^{**} = \phi^*$. As in Neyman–Pearson, $E_{\mu_1}\phi^* \geq E_{\mu_1}\phi$ for every ϕ with the side conditions, for all μ_1.
A test with $\max_{\Omega_0} E_\theta\phi \leq \alpha \leq \min_{\Omega_1} E_\theta\phi$ is **unbiased**; for smooth power functions this implies the side conditions, so the two-sided test is **UMP unbiased**.

9.3 Nonparametric tests and asymptotic comparison

Now let $X_1, \ldots, X_n$ come from a continuous location model $f(x-\theta)$, f symmetric about 0, strictly increasing cdf F; θ is the median.

Example 9.8 (Sign test)

For $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$, $S_n = \sum_i \mathbf{1}\{X_i > \theta_0\} \sim B(n, p(\theta))$, $p(\theta) = 1 - F(\theta_0 - \theta)$, and H_0 is $p = \frac{1}{2}$. The test

$$\phi_S = \begin{cases} 1, & S_n \geq c+1, \\ \gamma, & S_n = c, \\ 0, & S_n \leq c-1, \end{cases} \qquad \sum_{k=c+1}^{n}\binom{n}{k}2^{-n} + \gamma\binom{n}{c}2^{-n} = \alpha,$$

has size α for every f. With $X_i = Z_i + \theta$, $E_\theta\phi_S$ increases in θ (ϕ_S is nondecreasing in each x_i), so ϕ_S also has level α for $H_0 : \theta \leq \theta_0$: the **sign test**.

A test whose size does not depend on the population form is **nonparametric** (distribution-free); its cost in efficiency is measured by the sample size for given power.

Example 9.9 (Power of the sign test)

By the normal approximation the sign test rejects when $S_n \geq np(\theta_0) + \sqrt{n}\sigma(\theta_0)z_\alpha$, $p(\theta_0) = \frac{1}{2}$, $\sigma^2(\theta) = p(\theta)(1-p(\theta))$; at $\theta_1 > \theta_0$

$$\gamma_n(\theta_1) \approx 1 - \Phi\Big(-\sqrt{n}\frac{p(\theta_1)-p(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)}z_\alpha\Big) \to 1.$$

For $\theta_{1n} = \theta_0 + K/\sqrt{n}$, $p(\theta_{1n}) - p(\theta_0) \approx (\theta_{1n}-\theta_0)f(0)$, and

$$\gamma_n(\theta_{1n}) \approx 1 - \Phi(-2\sqrt{n}(\theta_{1n}-\theta_0)f(0) + z_\alpha).$$

so power γ needs $n \approx (2f(0))^{-2}\big(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n}-\theta_0}\big)^2$.

Theorem 9.4 (Approximate power and sample size)

For $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$, let a test reject when $\sqrt{n}(T_n - \mu(\theta_0))/\sigma(\theta_0) \geq t_n$, with size α, where

$$\sqrt{n}\,(T_n - \mu(\theta))/\sigma(\theta) \xrightarrow{d} N(0,1)$$

(uniformly for θ near θ_0) and μ, σ are continuously differentiable. Then:

(a) for $\theta_{1n} = \theta_0 + K/\sqrt{n}$, $\gamma_n(\theta_{1n}) \approx 1 - \Phi\big(-\sqrt{n}(\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)/\sigma(\theta_0) + z_\alpha\big)$;

(b) the sample size needed for power γ at θ_{1n} is

$$N(T; \gamma, \theta_{1n}) \approx \Big(\frac{\dot{\mu}(\theta_0)}{\sigma(\theta_0)}\Big)^{-2}\Big(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n} - \theta_0}\Big)^2.$$

Proof. Size α and asymptotic normality at θ_0 give $t_n \approx z_\alpha$. Under θ_1,

$$P_{\theta_1}\Big(\sqrt{n}\frac{T_n - \mu(\theta_1)}{\sigma(\theta_1)} \geq -\sqrt{n}\frac{\mu(\theta_1) - \mu(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)}t_n\Big) \approx 1 - \Phi\Big(-\sqrt{n}\frac{\mu(\theta_1) - \mu(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)}z_\alpha\Big).$$

For θ_{1n}, $\mu(\theta_{1n}) - \mu(\theta_0) \approx (\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)$ and $\sigma(\theta_{1n}) \to \sigma(\theta_0)$: (a). For (b) solve $\sqrt{N}(\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)/\sigma(\theta_0) - z_\alpha = z_{1-\gamma}$. ■

Definition (Asymptotic relative efficiency of tests)

For tests based on T_{1n}, T_{2n} satisfying Theorem 9.4,

$$ARE(\{T_{1n}\}, \{T_{2n}\}) = \lim \frac{N(T_2; \gamma, \theta_{1n})}{N(T_1; \gamma, \theta_{1n})} = \frac{(\dot{\mu}_1(\theta_0)/\sigma_1(\theta_0))^2}{(\dot{\mu}_2(\theta_0)/\sigma_2(\theta_0))^2}.$$

Example 9.10 (Sign test vs t test)

For $\frac{1}{\sigma}f(\frac{x-\mu}{\sigma})$, the sign test needs $N(S; \gamma, \mu_{1n}) \approx (2f(0)/\sigma)^{-2}(\frac{z_\alpha + z_{1-\gamma}}{\mu_{1n} - \mu_0})^2$; the t statistic has $\dot{\mu}/\sigma = 1/\sqrt{\operatorname{Var} X_1}$, $\operatorname{Var} X_1 = \sigma^2 \int z^2 f(z) dz$. So

$$ARE(\{S_n\}, \{T_n\}) = 4f(0)^2 \int z^2 f(z)\, dz,$$

as for median vs mean (Example 8.16):

	$N(\mu, \sigma^2)$	$L(\mu, \sigma)$	$DE(\mu, \sigma)$
$ARE(S, T)$	$2/\pi = 0.637$	$\pi^2/12 = 0.822$	2

Using ranks as well as signs gives a much more efficient nonparametric test.

Definition (Signed rank statistic)

With $R(|X_i - \theta_0|) = 1 + \sum_j \mathbf{1}\{|X_j - \theta_0| < |X_i - \theta_0|\}$, the **signed rank statistic** is

$$W_n = \sum_{i=1}^{n} \operatorname{sgn}(X_i - \theta_0)\, R(|X_i - \theta_0|),$$

and $W_n^+ = \sum_i \mathbf{1}\{X_i > \theta_0\} R(|X_i - \theta_0|)$ sums the ranks of positive deviations.

Theorem 9.5 (Null distribution)

Under $H_0 : \theta = \theta_0$:

(a) $W_n \overset{d}{=} \sum_{j=1}^{n} j\, S(j)$, with $S(1), \dots, S(n)$ i.i.d., $P(S(j) = 1) = P(S(j) = -1) = \frac{1}{2}$;

(b) $EW_n = 0$, $\operatorname{Var} W_n = \sum j^2 = n(n+1)(2n+1)/6$, and $W_n / \sqrt{\operatorname{Var} W_n} \xrightarrow{d} N(0,1)$.

Proof. (a) Let $S(i) = \operatorname{sgn}(X_i - \theta_0)$, $R(i) = R(|X_i - \theta_0|)$. By symmetry

$$P(|X_1 - \theta_0| \le x,\ \operatorname{sgn}(X_1 - \theta_0) = \pm 1) = \tfrac{1}{2} P(|X_1 - \theta_0| \le x),$$

so $|X_i - \theta_0| \perp \operatorname{sgn}(X_i - \theta_0)$; hence $S \perp R$, with i.i.d. fair signs. With inverse ranks $R^{-1}(j)$,

$$W_n = \sum_i S(i)R(i) = \sum_{j=1}^{n} S(R^{-1}(j))\, j.$$

For each permutation π, by $S \perp R$ and exchangeability of S,

$$P(S(R^{-1}(j)) = s_j\ \forall j,\ R = \pi) = P(S(\pi^{-1}(j)) = s_j\ \forall j) P(R = \pi) = P(S(j) = s_j\ \forall j) P(R = \pi).$$

Sum over π.

(b) Moments from (a). With $\sigma_n^2 = \sum j^2$, $Z_n = W_n / \sigma_n$ has cgf

$$\sum_{j=1}^{n} \log \frac{e^{jt/\sigma_n} + e^{-jt/\sigma_n}}{2} = \sum_j \log\Big(1 + \frac{j^2 t^2}{2\sigma_n^2} + \dots\Big) = \frac{t^2}{2} + O\Big(\sum_j \frac{j^4}{\sigma_n^4}\Big),$$

and $\sum j^4 / \sigma_n^4 = O(n^5/n^6) \to 0$. ■

Theorem 9.6

(a) $W_n = 2W_n^+ - n(n+1)/2$.

(b) Under H_0, $W_n^+ \overset{d}{=} \sum_{j=1}^{n} j B_j$ with $B_j \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\frac{1}{2})$, so $EW_n^+ = n(n+1)/4$, $\operatorname{Var} W_n^+ = n(n+1)(2n+1)/24$, and $(W_n^+ - n(n+1)/4) / \sqrt{n(n+1)(2n+1)/24} \xrightarrow{d} N(0,1)$.

Proof. (a) $\operatorname{sgn}(X_i - \theta_0) = 2\mathbf{1}\{X_i > \theta_0\} - 1$ (no ties), and $\sum R(i) = n(n+1)/2$.

(b) By (a) and Theorem 9.5(a), $W_n^+ \overset{d}{=} \sum j(S(j)+1)/2$. ■

The null law of W_n is free of f; large W_n^+ supports $\theta > \theta_0$. The **signed rank test** is

$$\phi_{SR} = \begin{cases} 1, & W_n^+ \ge c+1, \\ \gamma, & W_n^+ = c, \\ 0, & W_n^+ \le c-1, \end{cases} \qquad E_{\theta_0} \phi_{SR} = \alpha,$$

or for large n: reject when $(W_n^+ - n(n+1)/4) / \sqrt{n(n+1)(2n+1)/24} \ge z_\alpha$.

Example 9.11

$n = 10$, differences $1.8, -0.4, 2.5, 0.9, -1.1, 3.2, 0.6, 1.5, -0.2, 2.1$: absolute ranks $7, 2, 9, 4, 5, 10, 3, 6, 1, 8$, $W^+ = 47$. Null mean 27.5, SD 9.81, $z = 1.99 > 1.645$: reject at 5%. The sign test ($S = 7$, $P(S \ge 7) = 0.172$) does not.

Theorem 9.7 (Walsh averages)

(a) $W_n^+ = \sum_{1\le i\le j\le n} \mathbf{1}\{X_i + X_j > 2\theta_0\}$.

(b) The power function of ϕ_{SR} increases in θ; hence $\max_{\theta\le\theta_0} E_\theta\phi_{SR} = \alpha$, and ϕ_{SR} is a level-α test of $H_0 : \theta \le \theta_0$ vs $H_1 : \theta > \theta_0$.

Proof. (a) With $Y_i = X_i - \theta_0$ sorted by absolute value, $|Y_{(1)}| < \cdots < |Y_{(n)}|$,

$$W_n^+ = \sum_i \mathbf{1}\{Y_{(i)} > 0\}\cdot i = \sum_i \mathbf{1}\{Y_{(i)} > 0\} + \sum_i \sum_{j<i} \mathbf{1}\{Y_{(i)} > 0\}.$$

For $j < i$, $Y_{(i)} > 0$ iff $Y_{(i)} + Y_{(j)} > 0$, so $W_n^+ = \sum_{i\le j} \mathbf{1}\{Y_i + Y_j > 0\}$.

(b) By (a), ϕ_{SR} is nondecreasing in each X_i; with $X_i = Z_i + \theta$, $E_\theta\phi_{SR}$ is nondecreasing in θ. ■

Behaviour under alternatives (stated without proof):

Theorem 9.8

Let $\mu(\theta) = \frac{1}{2}P_\theta(X_1 + X_2 > 2\theta_0)$ and $\sigma^2(\theta) = \mathrm{Cov}_\theta\left(\mathbf{1}\{X_1 + X_2 > 2\theta_0\}, \mathbf{1}\{X_1 + X_3 > 2\theta_0\}\right)$. Then $E_\theta W_n^+ \approx n^2\mu(\theta)$, $\mathrm{Var}_\theta W_n^+ \approx n^3\sigma^2(\theta)$, and $(W_n^+ - E_\theta W_n^+)/\sqrt{\mathrm{Var}_\theta W_n^+} \xrightarrow{d} N(0,1)$.

Theorem 9.9 (Power of the signed rank test)

For $\theta_{1n} = \theta_0 + K/\sqrt{n}$,

$$\gamma_{\phi_{SR}}(\theta_{1n}) \approx 1 - \Phi\Big(-\sqrt{n}(\theta_{1n} - \theta_0)\frac{\dot\mu(\theta_0)}{\sigma(\theta_0)} + z_\alpha\Big), \qquad \dot\mu(\theta_0) = \int f^2(x)\,dx, \quad \sigma^2(\theta_0) = \frac{1}{12},$$

and the required sample size is $N(W^+;\gamma,\theta_{1n}) \approx \left(\sqrt{12}\int f^2\right)^{-2}\left(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n}-\theta_0}\right)^2$.

Proof. Apply Theorem 9.4 to $T_n = W_n^+/n^2$. With $Z \sim F$,

$$\mu(\theta) = \frac{1}{2}P(Z_1 + Z_2 > 2(\theta_0 - \theta)) = \frac{1}{2}\int\{1 - F(2\theta_0 - 2\theta - x)\}f(x)\,dx,$$

so $\dot\mu(\theta_0) = \int f^2$. At θ_0, $\mathbf{1}\{Z_1 + Z_2 > 0\}$ and $\mathbf{1}\{Z_1 + Z_3 > 0\}$ are conditionally independent given Z_1, each with mean $F(Z_1) =: U \sim U(0,1)$, so $\sigma^2(\theta_0) = \mathrm{Var}\,U = \frac{1}{12}$. ■

Example 9.12 (Signed rank test vs t test)

Since $\int(\frac{1}{\sigma}f(\frac{x}{\sigma}))^2dx = \frac{1}{\sigma}\int f^2$, Theorem 9.9 and Example 9.10 give

$$ARE(\{W_n^+\},\{T_n\}) = 12\Big(\int f^2(z)\,dz\Big)^2\int z^2 f(z)\,dz.$$

	$N(\mu,\sigma^2)$	$L(\mu,\sigma)$	$DE(\mu,\sigma)$
$ARE(W^+,T)$	$3/\pi = 0.955$	$\pi^2/9 = 1.097$	1.5

Under normality the loss is below 5%; for heavier tails the signed rank test wins, with distribution-free size.